Isometric Standard Project Verification & Storage Site Validation Report

For

Charm Industrial

Client	**Isometric**
Company / GHG Removal Supplier	**Charm Industrial**
Facility or Project Name	**Charm Bio-Oil Geologic Storage**
Facility Address	**Bio-oil Production:** **AE Cote-Nord Canada Bioenergy Inc (AECN)** Port Cartier, Quebec, Canada **Charm Miniforge** Fort Lupton, CO, USA **Bio-Oil Pre-Processing:** **Charm Industrial** El Dorado, KS, USA **Bio-oil Injection:** **Charm Industrial** Evangeline Parish, LA, USA
Greenhouse Gas Statement ID/Gas Statement Version ID	**ggs_1K3VXDCFV1S06CJ8/ gss_1K5HJ0HE51S0GJ2N**
Net Greenhouse Gas Removals	**135.39 tonnes CO_{2e}**
Removal Method	**Bio-Oil Geological Storage**
Removal Period	**August 1, 2025 – August 15, 2025**
Verification Opinion	**Positive (fully meets criteria)**
Verifier/Validation and Verification Body	**Tim Hansen** **Steven Qiu** **350Solutions, Inc.**

ISSUED:

SEPTEMBER 23, 2025

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ISOMETRIC STANDARD VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Steven Qiu
Removal Method: Bio-Oil Geological Storage		
Report Date: September 23, 2025		
Document No: 350VR-Charm-ISO2504.06		
Rev: v1.2		

** Primary contact/lead author*

1. KEY PROJECT DATA

TABLE 1. VERIFICATION SUMMARY FOR CHARM INDUSTRIAL

Title	Charm Bio-Oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Charm Industrial Miniforge 2360 S Rollie Ave Fort Lupton, CO, USA 80621 40.076570, -104.804680 Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection: Charm Industrial Evangeline Parish, LA 30.916251, -92.297908
Reporting Period	August 1, 2025 – August 15, 2025
Greenhouse Gas Statement ID	Final Revised: ggs_1K3VXDCFV1S06CJ8 (Version ID gss_1K5HJ0HE51S0GJ2N) (19SEP) Original Reviewed: ggs_1K3VXDCFV1S06CJ8 (Version ID gss_1K5AA6BQ21S0P4YP) (16SEP)
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (Quebec mill waste supplier); Altitude Forestry (CO forestry fire prevention waste supplier) Geologic Storage Management: Charm Industrial Regulatory Agency (Injections / Storage): Louisiana Department of Energy & Natural Resources
Legal Ownership of the Credits	Charm Industrial
Removals in Reporting Period	135.39 tonnes

Removal Method	Bio-Oil Geological Storage
Verification Report Date	September 23, 2025
Verification Type & Criteria	Isometric Standard Verification for ongoing deliveries of carbon removal credits by Charm Industrial in accordance with the documents used during project validation: • Isometric Standard v1.7 • Bio-Oil Geological Storage Protocol v1.1 • Biomass Feedstock Accounting Module v1.2 • Bio-oil Storage in Permeable Reservoirs v1.1 • Embodied Emissions Accounting v 1.0 • Energy Use Accounting v 1.2 • Transportation Emissions Accounting v1.1

1.1. VERIFICATION & STORAGE SITE VALIDATION SUMMARY

350Solutions was contracted to perform a project verification of the greenhouse gas (GHG) statement of Charm Industrial's bio-oil geological storage process for the referenced reporting period. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast, which focused on the use of third-party produced bio-oils and was issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. An additional validation of the Charm Kansas Bio-oil Geologic Storage project for use of bio-oil produced by Charm was completed on October 18, 2024, for the same crediting period. These validations remain valid for this verification reporting period. Further details and findings can be found in the validation report (https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation). A new injection and storage location in Evangeline Parish, Louisiana was utilized for the first time during this reporting period, and storage site validation is included in this report.

350Solutions affirms that Charm has the appropriate equipment, procedures, and protocols in place and is documented to quantify GHG removal through the injection of bio-oil into permeable reservoirs via a permitted Class V well in accordance with the requirements of the applicable Isometric standards, protocols and modules.

The carbon removals for the reporting period are provided in Table 2. These values have been verified by 350Solutions.

TABLE 2. VERIFIED TOTAL REMOVALS BY CHARM INDUSTRIAL FOR REPORTING PERIOD ENDING AUGUST 15, 2025

Removal ID	Date removal started	Date removal completed	Discounted CO_2e net removed, rounded (kg)
rmv_1K3MA3MX01S0823S	2025-08-14	2025-08-14	39380
rmv_1K3HJKAQC1S0YP7J	2025-08-11	2025-08-11	2960
rmv_1K3PJJBN31S0KDER	2025-08-15	2025-08-15	33620
rmv_1K3HW668J1S061A9	2025-08-13	2025-08-13	59430
TOTAL			135390

The quantity of **135.39 tonnes** of net carbon removals has been verified with a reasonable level of assurance and are eligible for crediting under the referenced Isometric Standard and Protocol.

350Solutions Inc. declares that it is an impartial verification body, approved by Isometric for verifications in the required technical area, free from any conflicts of interest, capable, and qualified to complete this verification for the current reporting period according to the *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2.1*, *Bio-oil Storage in Permeable Reservoirs Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*.

Verifier Information		
Verification Body	**Verifier**	**Verification ID No.**
350Solutions, Inc.	Tim Hansen	VR-Charm-ISO2504.06

Timothy A. Hansen Digitally signed by Timothy A. Hansen Date: 2025.09.23 14:21:01 -04'00'

Signed: Tim Hansen (Verifier)

Steven Qiu Digitally signed by Steven Qiu Date: 2025.09.23 11:48:02 -07'00'

Steven Qiu (QA)

2. VERIFICATION & STORAGE SITE VALIDATION DETAILS

2.1. VERIFICATION OBJECTIVES

A series of verification activities was conducted by 350Solutions to independently verify the claimed removals of the GHG statement submitted by Charm Industrial for operations conducted at the Charm Bio-oil Geologic Storage facility for the referenced Reporting Period. The verification was conducted following the specifications of *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2*, *Bio-oil Storage in Permeable Reservoirs Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*.

2.2. STORAGE SITE VALIDATION OBJECTIVES

Charm has initiated use of a new bio-oil injection and storage site for this reporting period. 350Solutions has conducted validation activities to ensure that the storage site, its operations, associated emissions, monitoring, and reporting procedures comply with the requirements of the *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, and Bio-oil Storage in Permeable Reservoirs Module v1.1.*

2.3. RESPONSIBILITIES

The verification is based on the provision and fair presentation of a complete and accurate GHG Statement submitted by Charm Industrial to the Isometric registry. It is Charm's responsibility to ensure a fair, accurate, and complete presentation of the GHG Statement and Project Design Document are provided. This is confirmed and verified independently by 350Solutions based on supporting evidence in accordance with the Isometric criteria (Table 1, Section 3.2).

2.4. LEVEL OF ASSURANCE & MATERIALITY

The verification was conducted with a reasonable level of assurance. Any discrepancies between the claimed removals of the project and the removals estimated by the verification process for the reporting period are less than the materiality threshold of 5%.

2.5. VERIFICATION TEAM DETAILS

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020:2012 for completion of ISO 14034:2016 Technology Verifications. 350Solutions is headquartered in Raleigh, North Carolina, USA. 350Solutions Technical Lead for the Charm verification is Tim Hansen. Mr. Hansen is a registered Professional Engineer with over 25 years of experience in the evaluation and verification of novel cleantech and carbon technologies. Mr. Hansen has led verifications of carbon removal technologies for multiple registries, including Isometric, and has experience in the development and testing of biomass conversion technologies, including gasification and pyrolysis. The verification report was provided QA by Steven Qiu, a Carbon Removal Verification Engineer, with experience developing verifying carbon removal technologies, including biochar and biomass terrestrial storage.

2.6. AUDIT PROCESS FOLLOWED

The specific verification and storage site validation activities conducted are summarized in Table 3. The list of all files that were reviewed is provided in Appendix 1. The list of findings from the verification can

be found in Appendix 2. Audit results are documented using 350's Compliance Checklist, which includes all requirements from the Isometric standard, Bio-oil Geologic storage protocol, and referenced modules.

TABLE 3. VERIFICATION & STORAGE SITE VALIDATION ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
September 2 -18, 2025	Review of reporting period modifications and changes	• Review of GHG Statement and report and guidance from Charm	Final V 1.10 Charm Industrial Bio Oil Geologic Storage PDD.docx H1 August Removals Isometric GHG Statement Report v1.6 VVB Guidance.docx ggs_1K3VXDCFV1S06CJ8 (Version ID gss_1K5HJ0HE51S0GJ2N) (19SEP) ggs_1K3VXDCFV1S06CJ8 (Version ID gss_1K5AA6BQ21S0P4YP) (16SEP)
September 18-22, 2025	Review of current Validation and GHG Statement Applicability	• Review of Validation Report • Review of prior Verification Report and findings • Identification of major and minor process steps, measurements, and calculations and changes in RP • Review of Standards/Protocols/Modules required for Verification • Generation of a checklist of verification issues and findings and opportunities for improvement	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1 Final V 1.10 Charm Industrial Bio Oil Geologic Storage PDD.docx H1 August Removals Isometric GHG Statement Report v1.6 VVB Guidance.docx ggs_1K3VXDCFV1S06CJ8 (Version ID gss_1K5HJ0HE51S0GJ2N) (19SEP) ggs_1K3VXDCFV1S06CJ8 (Version ID gss_1K5AA6BQ21S0P4YP) (16SEP) Isometric Standard V1.7.2, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2, Bio-oil Storage in Permeable Reservoirs Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1, and Energy Use Accounting v1.2
September 18-22, 2025	Review of new Storage Site information and Validation of site	• Review of revised PDD and supporting information regarding Basco Storage site • Review of Bio-oil Storage in Permeable reservoir module • Review of supporting documentation including UIC permit • Generation of validation checklist and issuance of log of storage site validation findings	Final V 1.10 Charm Industrial Bio Oil Geologic Storage PDD.docx Isometric Standard V1.7.2, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2, Bio-oil Storage in Permeable Reservoirs Module v1.1 VVB Guidance.docx See Appendix 1 for file log
September 4, 2025	Site visit to Basco storage site	• On-site review of storage site equipment, instrumentation, measurements, processes and operations • Observation of bio-oil injections and collection of monitoring data • Review of supporting data and operating procedures for new site • Development of request for additional information and initial validation findings	On-site review of standard operating procedures, EHS procedures, monitoring plan On-site and post site visit review of data collected during site visit

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
September 18-22, 2025	Verification of Removals	• Verify changes to the system • Review of measurements and calculations • Review of durability/permit requirement monitoring data • Review of GHG statement • Identification of initial findings and follow-up request for information • Write verification report	File List provided in Appendix 1. Initial Findings in Appendix 2.

3. TECHNICAL ASPECTS

3.1. PROJECT DETAILS

Charm obtained bio-oil from two sources during this reporting period. Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill. In addition, Charm produces bio-oil at its pyrolysis facility in Fort Lupton, CO. Production of bio-oil for sequestration and carbon removal credit generation began in August of 2024, with the initial installation of 4 pyrolyzers, with plans for up to 18 pyrolyzers total to be installed. Charm's pyrolyzers utilize wood chips, placed into a high temperature pressurized reactor in a low oxygen environment to reduce the wood chips into a combination of products, including syngas and light hydrocarbon gases, pyrolysis oils, biochar, and wood vinegar. All hot pyrolysis gases from the reactor are quenched with water to produce the oil and wood vinegar products. Excess gases are routed to a combustion device for emissions control, with the heat produced from the combustion of excess gas utilized for drying the biomass from 40%-50% moisture content to 10% moisture content before introduction into the pyrolyzer.

Bio-oil from AECN and aqueous fraction product from Charm pyrolyzers are shipped by tanker truck, rail car, or tote tanks on flatbed to a Charm facility in El Dorado, Kansas. At this facility, Charm pretreats the bio-oil through the addition and sparging of waste brine (salt) and/or liquid caustic soda (LCS) as needed to adjust the pH of the bio-oil and aqueous product to meet the requirements of the injection facility permit for storage in permeable reservoirs as specified by the Louisiana Department of Energy & Natural Resources (LDENR). Charm-produced bio-oil is not pretreated when injected at the Basco injection site in Louisiana.

Charm Bio-oil product and the treated AECN oil and Charm aqueous product are trucked to Charm's underground injection and storage facility located in Evangeline Parish, Louisiana for storage into a geologic reservoir. The bio-oil products are blended in on site storage tanks through recirculating pumps and injected as soon as practicable. Samples of the blended bio-oil are collected prior to injection and analyzed for bio-oil characteristics to ensure compliance with permit requirements. Charm also monitors all injection parameters as required by the Underground Injection Control (UIC) permit.

3.2. PROTOCOLS AND MODULES APPLIED

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard v1.7 [1]
- Bio-Oil Geological Storage Protocol v1.1 [2]
- Biomass Feedstock Accounting Module v1.2 [3]
- Bio-oil Storage in Permeable Reservoirs v1.1 [4]
- Embodied Emissions Accounting v 1.0 [5]
- Energy Use Accounting v 1.2 [6]
- Transportation Emissions Accounting v 1.1 [7]

Charm utilizes Book and Claim Units (BCUs) to substitute for some, or all, or project diesel fuel usage and purchases Renewable Energy Certificates (RECs) for some or all its electricity consumption in accordance with the Energy Use and Transportation Modules.

3.3. APPLICABILITY OF APPLIED PROTOCOL AND MODULES

The Charm project takes mixed woody waste or forestry thinnings from wildfire prevention activities, converts it to bio-oil, which they pretreat and inject into geologic storage formations. The biomass feedstocks were assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and were found to be eligible. As the biomass feedstocks have been deemed eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in a geologic formation makes both the Bio-oil Geological Storage Protocol and Bio-oil Storage in Permeable Reservoirs Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.4. PROJECT BOUNDARY

The boundaries of the project include the supply of wood waste to the biomass processing sites (AECN and Charm Fort Lupton), the conversion of the biomass to bio-oil via AECN's pyrolysis process and Charm's pyrolysis process, transport of the bio-oil to Charm for preprocessing at its El Dorado, Kansas facility, completion of pretreatment, shipment to the Charm Louisiana injection and storage site, and injection of the bio-oil in the geologic reservoir. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary.

3.5. CHANGES FROM PREVIOUS VERIFICATIONS

Compared to prior verifications, Charm has made modifications to the process, data collection, and calculation procedures which have been identified in the GHG statement and supporting documentation. The following modifications in project activities were made, in addition to updates to delivery data and monitoring data for the reporting period. All were reviewed and deemed acceptable under the Isometric requirements:

3.5.1. NEW BASCO INJECTION & STORAGE SITE

Charm has obtained rights for use of a former oil and gas brine disposal well located in Evangeline Parish, Louisiana for injection and storage of bio-oil. The well is a former brine disposal well (Basco 6 EW Well, Louisiana DNR Well Serial Number 974025) located in Evangeline Parish, LA. It was originally drilled in 2011, and while active, injected nearly 20,000 barrels of produced water before transitioning to orphan well status in 2016. The well injects through perforations at a depth of 3534 feet into the Lower Miocene sandstone formation. The storage formation has been characterized and assessed for the performance required to ensure durable storage of injected bio-oil. The injection point is far below the lowermost underground source of drinking water (USDW) which is located at 1170 feet below surface

with the storage formation separated from drinking water sources by a confining layer of non-transmissive caprock. The well injection permit was obtained by Charm and issued with an effective date of August 8, 2025, with specifications for injection of bio-oil via the class V well into this formation.

Charm completed assessments of the well and formation storage capacity and system integrity as part of the permitting process. Charm determined that the pressure required to pump bio-oil into the formation would be consistent with a typical injection formation (normally pressured or below) without exceeding maximum allowable injection pressure (defined in the permit as 455 psi, with pressure testing completed at 1000 psi during permitting), and that the formation brine density (~1.02 g/cm^3) is sufficiently lower than that of bio-oil (>1.2 g/cm^3). This ensures that Charm can pump bio-oil under the regulatory requirements and that upon entry into the formation, the bio-oil will sink. In addition, Charm performed a mechanical integrity test and material corrosion inspection to ensure casing integrity and performed an advanced cement analysis to ensure hydraulic seal from the formation and the wellbore. The wellbore was proven to be in good mechanical condition, and the formation is of sufficient injectivity and storage size.

Once successfully pre-treated to ensure UIC/LDENR compliance, bio-oil is trucked to Charm's Basco 6 injection well in Evangeline Parish, LA for injection and permanent storage. The bio-oil may be stored in tanks prior to injection into the sub-surface or injected directly from tanker trucks. When a batch of bio-oil is ready for injection, Charm injects a small amount of brine into the formation to ensure pressure and formation access is achieved. Once injection into the subsurface has been established with brine, Charm switches to the injection of bio-oil. As required by the UIC permit, Charm continuously monitors the injection rate and surface injection pressures at the wellhead. Automated shutdowns are in place to ensure compliance with regulation. Bio-oil is continuously pumped through the injection pump until the tank(s) containing the injection batch is empty. After all pre-processed bio-oil is injected, Charm completes additional brine flush to clear residual oil from the injection lines, pump, and wellbore to ensure all bio-oil is delivered into the formation. After this brine flush is complete, the wellbore is shut-in, and pressure (well and annulus) continuously monitored in preparation for the next injection. An ongoing monitoring program as required by and in compliance with the UIC permit and Isometric storage module has been established and is detailed in the PDD and supplemental documents and reports.

3.5.2. ADDITIONAL MODIFICATIONS

- Embodied emissions have been adjusted to include both include both the existing KS and new LA site equipment, sitework, etc. The amortization is based on the projected gross removals that will be completed at the Basco site. The total pool of embodied emissions is divided by the total expected gross removals and that factor is then multiplied by the total gross removals in the period.
- All Charm oily fraction bio-oil was produced during Production Period 8, production emissions were previously verified under the prior reporting period (Jul1-31, 2025). RECs and BCUs were already allocated during the June/July period, so no additional allocations for CO were made for this Reporting Period.
- All Charm aqueous fraction was produced prior to Production Period 7, so the current reporting period aqueous fraction removals are not burdened with any production emissions as all

production emissions prior to Production Period 7 were allocated to oily fraction and biochar only and fully accounted for in prior bio-oil storage reporting periods.

- The following operational changes to the material flow have been made:
    - Aqueous fraction from Charm pyrolyzers is trucked from CO to KS where the oil is sparged and buffered. The "sparging batch" is then offloaded into a tanker truck where it is transported to LA and injected
    - AECN is trucked from QC to KS where the oil is sparged. The "sparging batch" is then offloaded into a tanker truck where it is transported to LA and injected
    - Totes of the Charm oil fraction are trucked from CO directly to LA where they are emptied into a holding tank and recirculated with AECN oil to homogenize the injectate prior to injection
    - The Mass of Bio-Oil Injected for each batch is computed by subtracting the mass of a tanker after offloading bio-oil from the mass of the fully laden truck as documented by scale tickets from certified truck scales
        - In the case of Charm oil/AECN blends, the mass of Charm oil (minus loss) is added to the mass of AECN biooil from the tanker delivering the processed AECN oil used in a given batch
- The UID for RECs in the Inventory Tracker has been changed to the last 4 digits of the certificate serial number.
- There are no Waste Disposals for this reporting period
- Note that Charm is continuing to transition additional parts of its reporting to the Isometric Certify platform. As this occurs, certain components may appear partially complete or incomplete in Certify. However, the data has been submitted separately and verified by 350 as compliant. For this reporting period, Monitoring and Feedstocks requirements are included in Certify, but entries appear incomplete. However, these items are fully addressed outside of Certify.

3.6. PROCESS & DATA IRREGULARITIES DURING REPORTING PERIOD

No significant data anomalies or irregularities occurred during the reporting period related to data collection and operations.

4. CREDITING DETAILS

4.1. PROJECT REPORTING PERIOD

This verification considers injections of bio-oil (oil and aqueous phases) by Charm during the period August 1, 2025 to August 15, 2025. The reporting period is based on the actual injection dates, with bio-oil production, transportation, monitoring, and related activities potentially occurring outside of this reporting period, but properly accounted for and allocated to the bio-oil injected within the reporting period. Note that injections occurred beginning August 11, after the issuance of the Basco site permit effective August 8, 2025.

4.2. CREDITING PERIOD

The 5-year crediting period for the Charm Kansas Bio-oil Geologic Sequestration Project began March 1, 2024, and continues for five years, ending February 28, 2029. All claimed removals in this reporting period fall within the current crediting period.

5. GHG STATEMENT VERIFICATION & STORAGE SITE VALIDATION

5.1. ACCURACY OF THE STATEMENT

The values represented in the GHG statement have been evaluated, cross-referenced with the presented evidence, and confirmed, noting:

- The initial GHG Statement contained an error in weights for one bio-oil injection batch, which was corrected and modified the total GHG emissions slightly. A revised GHG Statement was submitted and verified.

Confirmation of the GHG statement has been based on the following efforts:

- Review and confirmation of calculations of total bio-oil and CO_2 equivalent injected and all process, operational, and embodied emissions, as well as all supporting evidence for the following removals:

rmv_1K3HJKAQC1S0YP7J	11 Aug 2025
rmv_1K3PJJBN31S0KDER	15 Aug 2025

- This includes for these removals:
  - Calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content, addition of consumables)
  - Review and calculation of average carbon content from raw data (lab reports)
  - Confirmation of transportation distances for tanker and rail car deliveries from AECN to El Dorado, KS and to Basco site, LA
  - Confirmation of weigh scale raw data for deliveries of bio-oil and injectate
  - Re-calculation of transportation associated emissions for such transport
  - Confirmation of raw data for fuel usage at Charm pre-treatment and injection facilities
  - Confirmation of electricity usage at Charm pretreatment and injection facilities
  - Confirmation of proper use and accounting of BCUs and RECs to offset diesel and electricity usage emissions, respectively
  - Review of Charm pyrolysis process emissions, supporting evidence, and calculations
- Full Review of Basco site equipment inventory, embodied emissions factors for selected equipment (telehandler, frac tank, pumps), and amortization process and calculations,
- Full Review of Basco site and injection process emissions
- Review of monitoring data as required for Basco site injection and post injection monitoring to ensure compliance with Class V permit.

5.2. VERIFICATION STORAGE SITE VALIDATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the greenhouse gas removal statement indicates that the claims fall within the 5% materiality threshold for greenhouse gas emissions and removals as required by the Isometric Standard:

☒ ***Meets the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Bio-oil Storage in Permeable Reservoirs Module v1.1 for Verification***

☐ ***Meets the requirements with minor revisions of Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Does Not Meet the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

5.3. VERIFICATION & STORAGE SITE VALIDATION RESULTS SUMMARY

A summary of verification results and compliance with relevant requirements of the *Isometric Standard, Biomass Geological Storage Protocol, Biomass Feedstock Accounting Module*, and *Bio-oil Storage in Permeable Reservoirs Module* are summarized below.

TABLE 4. VERIFICATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	Y	New site utilizes newly applicable protocol for Permeable Reservoirs. Acceptable – No additional changes since validation
2.5.1	*Boundaries*	Y	New Basco injection site incorporates new facility and modified boundary. Acceptable – No additional changes since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality – Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality – Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation and prior verification.
2.5.5	*Default Emission Factors, Proxies, and Models*	Y	Acceptable – New emissions and emission factors from Basco LA injection site accounted for and emissions factors selected are acceptable (embodied emissions, electricity). No additional significant changes were made since prior verifications. See Section 3.5 for summary of modifications to any calculation methods, all of which are acceptable.
2.5.11	*Life Cycle Assessment (LCA) Policies*	N	
2.5.6	*Common Calculation Factors*	N	

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.5.7	*Uncertainty*	N	
2.5.8	*Durability and Monitoring*	Y	New injection and storage site implemented during reporting period. All durability evidence and storage monitoring requirements validated to meet protocol requirements.
2.5.9	*Risk of Reversal*	Y	New storage site requires a modified risk assessment, which was provided. New storage module indicates a buffer pool of 5% is selected based on storage type and current science regardless of risk assessment. Acceptable – No additional changes since validation
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	Y	New storage site required additional environmental and social impact review. PDD modified with additional information. Acceptable – No additional changes since validation.
3.1	*Ownership*	N	Acceptable – No change since validation
3.2	*Documentation*	Y	Acceptable – New Basco storage site documentation provided as required and PDD updated. No additional changes since prior verification. All documentation properly updated for reporting period
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV Certify platform.
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete, and all removals have been verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No change since validation. New Basco site permitting included public comment and review.
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	New Basco storage site obtained Class V UIC well permit effective August 8, 2025. All permitting procedures for Basco site compliant. No additional compliance issues identified.
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	N	Acceptable – No change since prior feedstock validations
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since prior feedstock validations
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since prior feedstock validations
3.0	*Counterfactual Definition*	N	Acceptable – No change since prior feedstock validations
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since prior feedstock validations
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since prior feedstock validations

TABLE 5. BASCO STORAGE SITE VALIDATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
Bio-oil Storage in Permeable Reservoirs Module			
1.0	*Storage System Eligibility*	Y	Acceptable – new storage site meets eligibility requirements
2.1	*Storage Reservoir – Permitting*	Y	Acceptable – new storage facility meets permitting requirements and was issued a Class V UIC well specifically identifying bio-oil as injectant on August 8, 2025
2.2	*Storage Reservoir – Capacity*	Y	Acceptable – new storage facility has been demonstrated to have more than adequate storage capacity for planned removals based on testing data and porosity, permeability and site characterization
2.2	*Storage Reservoir – Integrity*	Y	Acceptable – new storage facility is properly contained by a confining layer that is free of transmissive faults and is not specified as susceptible to seismic activity
2.2	*Storage Reservoir – Integrity: Characterization*	Y	Acceptable – new storage facility was properly characterized and simulations provided demonstrating capacity. Reservoir fluids properly analyzed at baseline conditions.
2.2	*Storage Reservoir Characterization – Temperature, pH, conductivity, density*	Y	Acceptable – new storage site formation fluid evaluated as required. All required parameters measured and are within the required limits.
2.2	*Storage Reservoir Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – Not required
2.3	*Storage Reservoir Integrity – Well Construction*	Y	Acceptable – New well is a former oil and gas displaced brine injection well. Well was inspected, tested, and modified to ensure integrity during Charm operations. Construction materials are compatible with bio-oil. Well construction records are maintained.
3.1.1	*Storage Reservoir Integrity – Injection operations*	Y	Acceptable – New storage site injection limitations are defined in permit and site operations and controls and alarms are set to ensure compliance with permit conditions. Conditions and injections are instrumented and monitored as required by permit and protocol.
3.1.1	*Storage Reservoir Integrity – Injection operations – Sampling & Analysis*	Y	Acceptable – new storage site has procedures in place that have been implemented properly for sampling and analysis of each batch of injection material
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation
3.1.3	*Injection Pressure*	Y	Acceptable – Pressure is continuously monitored and reported quarterly (no monitoring reports required to be submitted during this period).
3.1.2	*Mechanical Integrity*	Y	Acceptable – New well mechanical integrity testing and well, casing and packer inspections completed prior to initiating injections and as part of permit approval. Corrosion monitoring done via semi-annual inspections
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change since validation
3.1.3	*Reversal Risk & Monitoring*	Y	Acceptable – groundwater sampling and AOR monitoring plan developed
3.3.1	*Attribution of Reversals*	Y	Acceptable

5.4. VERIFICATION & STORAGE SITE VALIDATION FINDINGS

5.4.1. CRITICAL FINDINGS AND EXCEPTIONS

Assessment of the greenhouse gas removal statement identified several initial findings (Appendix 2). All findings, primarily associated with missing supporting evidence, have been addressed and closed. Findings of note include the following:

ID No	Type	Finding / Issue	Required Action	Supplier Response	Resolution
1	Nonconformity - Omission	Weigh Scale ticket for 5-01 injection batch Aqueous delivery is not provided (empty scale ticket is provided, but not full)	Provide clarification or correct info (see Finding 2)	The full scale ticket for 5-01 is included — The original file name was "5-01 Scale In and Out" but was revised to "5-01 Scale In and Pre-load empty" for improved clarity as there are two scale tickets in the image: The pre-loaded empty mass (33600 lbs, weigh number 1408) and the full mass (61200 lbs, weigh number 2539). The pre-load empty is taken so there is a redundant unloaded datapoint, but the post-unload mass is preferred, so it is not used unless the post-unload mass is not available. The full mass reflects the same value that was included in the original ops notes and calculations for this removal.	Complete
2	Nonconformity - Error	It appears that the wrong empty weight was used for the bio-oil shipment truck weight for the 5-01 delivery of bio-oil. The weigh scale ticket shows 33600, but MRV Ops notes file shows 31620 (as does Certify - removal YP7J) - which is the aqueous truck empty weight and for a different truck and trailer. This drops the mass of bio-oil injected by 898 kg when corrected. Ensure that the total gross removals value which is used elsewhere is corrected and carried through (not material): Total embodied emissions (+2kg additional); transport emissions (empty truck) = -67kg additional; truck embodied (negligible)	Provide correct final weigh ticket for 05-01	The scale ticket showing 31620 lbs was included in error and is not the correct post-unload empty weight. The correct ticket providing the post-unload empty weight was identified and confirmed that the correct date and tractor/trailer combination are listed. The ticket weight is 33480 lbs and the weigh number is 3159. Ops notes and calculations in Certify have been updated to reflect the corrected value. This correction reduced the MOBI for Batch 5-01 by 843.68kg and reduced to total reported CDR by 280kg which is reflected in an updated GHG Statement Report.	Complete. GHG Statement updated 19SEP and verified. Revised total removals from 135.67 to 135.39 tonnes

9	Nonconformity	A description of the confining system or caprock is not complete. Although indications are that the system holds significantly more pressure than planned injection MASIP, and injections will be well below the USDW lowest point, details regarding formation and faults or transmissivity are unclear	Please provide a summary of the confining system and details, including	This is addressed in the document titled "Basco 6 Supplementary Subsurface Characterization" which is now included in the "VVB Feedback" subfolder of the "Additional Documentation" folder in addition to supporting evidence. The text of the document, minus attachment links that won't work for the public, is also reproduced as Appendix 9 of the updated PDD	Complete. Additional evidence supporting presence of confining layers provided in 'Basco 6 EW L Miocene Verification Requests.docx'. Note - if future reperforations are planned for injection at shallower depths, we would request similar data to the well bore GR, GR KUT, and other test results be provided for review and as evidence of confining layers in the planned injection region.
11	Nonconformity	Reservoir mineralogy and lithology descriptions are minimal	Please provide or direct to any more detailed descriptions of formation beyond the brief summary in the PDD	Additional mineralogy and lithology descriptions are also included in the Basco 6 Supplementary Subsurface Characterization document	Complete
13	Nonconformity	Protocols require validation that well materials designed for life of project and compatible with fluids / injectate and meet API, ASTM, ISO or comparable standards as applicable (i.e. corrosion resistance). Although well materials appear to be designed properly and are compatible, no evidence is provided on materials used not compliance with ASTM, API, or other standards	Please provide supporting information on construction materials used and required standards	This information is provided in the document titled "Material Selection and Service Life - Basco 6 EW 001" in the "VVB Feedback" subfolder of the "Additional Documentation" folder in addition to supporting evidence	Complete

5.4.2. FORWARD ACTION REQUESTS & RECOMMENDATIONS

Forward Action requests (FAR) and recommendations are also provided in Appendix 2 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of potential issues to be monitored by verifiers in future verifications. Open Forward Action Requests and opportunities for improvement are summarized below for future verification reference:

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS

ID No	Type	Finding / Issue	Required Action	Resolution
12	FAR	Site Formation characterization is required to be reviewed every ten years or when site conditions warrant, with comparison of pre-emplacement project assumptions to actual measured conditions and updates as needed	Please add the 10-yr review and review upon collection of measured conditions or observations differing from initial review to the Permit Compliance Plan in the Supplemental Document - Basco 6	Complete. Confirmed supplier statement that "This has been added to the Permit Compliance Plan."
5	FAR (prior)	The PDD should be updated to reflect GHG Statement calculation changes as noted in the referenced briefs		PDD Updated. Complete
6	FAR (prior)	Please plan to provide updated documentation on community engagement and stakeholder input activities by 12/31/2025		Review for future verifications
7	FAR (prior)	Please plan to provide updated documentation on grievance mechanism updates and any complaints/grievances received and responses (or confirmation of no issues) by 12/31/2025		Review for future verifications
8	FAR (prior)	Please ensure all waste disposal related emissions for future disposal of wastes which has been produced during this period are accounted for in future reporting periods as required. Note: Charm has properly accounted for all disposals so far and has all systems in place to do so going forward.		Not applicable for current verification. Review for future verifications

6. REVISION HISTORY

Original date of issue: September 20, 2025

Version	Date Issued	Noted Changes
V1.0 (Draft – Internal)	September 20, 2025	N/A Initial Version
V1.1 (Draft – External)	September 23, 2025	Minor edits from QA
V1.2 (FINAL)	September 23, 2025	Final Version – Minor Edits from Isometric and Charm; Closure of Findings List

References

[1] Isometric, “Isometric Standard v1.2.0.” Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/standard/1.2.0

[2] Isometric, “Bio-oil Geological Storage V1.1.1.” Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/bio-oil-geological-storage/1.1

[3] Isometric, “Biomass Feedstock Accounting V1.1.” Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/biomass-feedstock-accounting/1.1

[4] Isometric, “Biomass or Bio-oil Storage in Salt Caverns V1.0.4.” Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/salt-cavern-storage/1.0

[5] Isometric, “Embodied Emissions Accounting v1.0.3.” Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/embodied-emissions/1.0

[6] Isometric, “Energy Use Accounting v1.1.” Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/energy-use-accounting/1.1

[7] Isometric, “Transportation Emissions Accounting v1.1.” Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/transportation

See Appendix 1 and Table 3 for list of specific files reviewed during the verification.

APPENDIX 1: DOCUMENTS SUBMITTED FOR REVIEW

Name	File Topic
GHG Statement & PDD	
gss_1K5AA6BQ21S0P4YP.xlsx	\GHG Statement\
gss_1K5HJ0HE51S0GJ2N.xlsx	\GHG Statement\
Final V 1.10 Charm Industrial Bio Oil Geologic Storage PDD.docx	\PDD\
Final V 1.10 Charm Industrial Bio Oil Geologic Storage PDD_OLD.docx	\PDD\
PDD V1.10 Change Log.xlsx	\PDD\
Supplemental Doc 3 -- Basco 6 PDD Supplemental Document.xlsx	\PDD\
Response to Findings - Supporting Evidence	
1) S13CR On-Off Tool_103126985.pdf	\Charm Findings Responses\
2) WL Mech Set PLT Packer_102712316.pdf	\Charm Findings Responses\
Basco 6 EW L Miocene Verification Requests.docx	\Charm Findings Responses\
Findings-Charm_1AUG-15AUG2025_Final_v1.xlsx	\Charm Findings Responses\
Job59299-Water.pdf	\Charm Findings Responses\
Jun et al 2022.pdf	\Charm Findings Responses\
M25-025 Cross-section Analysis Report (Pre-tested AECN solution coupons) (1).pdf	\Charm Findings Responses\
Audit Data Package	
H1 August Removals Isometric GHG Statement Report v1.6.docx	\Additional Documentation\Additional Documentation\
VVB Guidance.docx	\Additional Documentation\Additional Documentation\
Community Engagement Events List (1).xlsx	\Additional Documentation\Community Engagement-20250919T201926Z-1-001\Community Engagement\
Basco 6 Embodied Emissions Inventory.xlsx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\
10500 Gal Water Tank.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
40_x8_ Conex Shipping Container.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
Diaphragm Seal for Pressure Transducers.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
Downhole Monitoring Assembly.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
Generac GP2500i.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
JCB Telehandler 512-56.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
Kubota L2501 Tractor.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
PC Pump Motor+VFD.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
Progressive Cavity Pump.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\

SRO System.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
Ultrasonic Flow Meter.docx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\LCA Docs\
acerotomold-tank_cb10500-142.pdf	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\Reference Docs\
AMETEK USG - Charm Industrial Special Quote for EW Seal # CHARM-EW-103124-BJ (1).pdf	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\Reference Docs\
Endress-Hauser_Prosonic_Flow_Inline_92F_EN.pdf	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\Reference Docs\
Proposal K20425CW.pdf	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\Reference Docs\
Quote #2061278337.pdf	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\Reference Docs\
SRO SYSTEM 09042024.xlsx	\Additional Documentation\Embodied Emissions Documentation-20250919T201929Z-1-001\Embodied Emissions Documentation\Reference Docs\
Screenshot 2025-07-16 at 2.51.36 PM.png	\Additional Documentation\Routes and Distances-20250919T201931Z-1-001\Routes and Distances\
Endress-Hauser_Prosonic_Flow_Inline_92F_EN.pdf	\Additional Documentation\Uncertainty Documentation-20250919T201932Z-1-001\Uncertainty Documentation\
Hawk 9500 General Description 2v14.pdf	\Additional Documentation\Uncertainty Documentation-20250919T201932Z-1-001\Uncertainty Documentation\
BOL 2195-T pickup 7.12.pdf	\AECN BOLs\AECN BOLs\
BOL 2196-T pickup 7.14.pdf	\AECN BOLs\AECN BOLs\
BOL 2197-T pickup 7.13.pdf	\AECN BOLs\AECN BOLs\
BOL 2198-T pickup 7.14.pdf	\AECN BOLs\AECN BOLs\
BOL 2199-T pickup 7.14.pdf	\AECN BOLs\AECN BOLs\
Recorded Well, Wellsite and Subsurface Storage Servitude Agreement _ Basco - File No. 663900, Evangeline Parish, 1_5_24(101912721.1) (1).pdf	\Basco Site & Well Permitting\
Charm Louisiana, LLC - Spill Prevention Control and Countermeasure Plan - Final to Client 8.27.25.pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
EHS Training Tracker.xlsx	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
EHS-0001 Charm Safety Handbook.pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
EHS-0019 Personal Protective Equipment Program.pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
EHS-0031 Louisiana EAP_20250820.pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
EHS-0042 Contractor Safety Orientation Reference Manual .docx	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
LDEQ HW1 AOR.pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
OSHA 300A Logs_2024.pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
PPE Matrix Visuals Revised 3.20.25.pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\

RTP™ Liquid RFO™ Biofuel Low Carbon Fuel Oil (LCFO) Fast Pyrolysis Bio Oil (FPBO) Biocrude - English (2023).pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
Wet Oil _ Dry Oil (WO_DO).pdf	\Basco Site & Well Permitting\EHS and Safety-20250919T202009Z-1-001\EHS and Safety\
B6 Jul MIT.pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
Basco Data Export - 9_4_25.xlsx	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
Basco Injection - Data Analysis.xlsx	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
CHARM INDUSTRIAL FIGURES.pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
CHARM LA BASCO 6 SWD 1 CAST PIPE INSPECTION.pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
Charm Louisiana LLC_Order No. IMD 2025-05 EW.pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
CURRENT WBD 01.28.25.pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
FOT Work Report - Basco 6 EW - December 2024.pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
LDENR Injection Reporting_ Permit Compliance Checklist and Monthly Reporting Template.xlsx	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
Re-Perf Work Permit.pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
SVBFS002815 copy of LC UPS (1).pdf	\Basco Site & Well Permitting\Permit Compliance and Non-CDR Monitoring-20250919T202011Z-1-001\Permit Compliance and Non-CDR Monitoring\
Basco 6 Flow Meter 1.jpeg	\Basco Site & Well Permitting\Site Meter Calibration Docs-20250919T202013Z-1-001\Site Meter Calibration Docs\
Basco 6 Flow Meter 2.jpeg	\Basco Site & Well Permitting\Site Meter Calibration Docs-20250919T202013Z-1-001\Site Meter Calibration Docs\
Downhole Pressure and Temp.pdf	\Basco Site & Well Permitting\Site Meter Calibration Docs-20250919T202013Z-1-001\Site Meter Calibration Docs\
prosenseets1001.pdf	\Basco Site & Well Permitting\Site Meter Calibration Docs-20250919T202013Z-1-001\Site Meter Calibration Docs\
prosensespt25transmitters.pdf	\Basco Site & Well Permitting\Site Meter Calibration Docs-20250919T202013Z-1-001\Site Meter Calibration Docs\
Production Period 8 Summary.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\
PP8 (7_1-7_31) Biomass Throughput.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Biomass Throughput\
Miniforge Biomass Transport Allocation -- Updated.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Biomass Transport\
PP8 (7_1-7_31) Vendor Delivery Emissions Calc Sheet.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Consumables Delivery\
Copy of Agfinity to Miniforge.png	\Charm BO MRV\Charm BO MRV\Production Period 8\Consumables Delivery\Vendor Distance Measurements\
Copy of Airgas to Miniforge.png	\Charm BO MRV\Charm BO MRV\Production Period 8\Consumables Delivery\Vendor Distance Measurements\
Copy of Miniforge to Ace.png	\Charm BO MRV\Charm BO MRV\Production Period 8\Consumables Delivery\Vendor Distance Measurements\
Copy of Miniforge to Buckeye.png	\Charm BO MRV\Charm BO MRV\Production Period 8\Consumables Delivery\Vendor Distance Measurements\
Diesel Consumption Calc 7_1_25-7_31_25.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Diesel\
Copy of 7.21.25.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Diesel\Transaction Docs\
Copy of 7_13_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Diesel\Transaction Docs\

Copy of 7_14_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Diesel\Transaction Docs\
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Copy of 7_28_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Diesel\Transaction Docs\
Copy of 7_30_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Diesel\Transaction Docs\
7_1_25-7_31_25 - 7_1_25-7_31_25.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Electricity\
7_1_25-7_31_25.csv	\Charm BO MRV\Charm BO MRV\Production Period 8\Electricity\
Date Range Screenshot.png	\Charm BO MRV\Charm BO MRV\Production Period 8\Electricity\
PP8 Miniforge Electricity Consumption Calculator.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Electricity\
Cap Ex MRV July.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Embodied Emissions\
SP&C July.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Embodied Emissions\
PP8 Gasoline Consumption Calculator.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Gasoline\
Gasoline Receipt.jpg	\Charm BO MRV\Charm BO MRV\Production Period 8\Gasoline\Gasoline Receipts\
Production Period 8 Output Summary.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Output Data\
Biochar C Content Testing.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Output Data\Testing\
WODO and Aqueous C Content.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Output Data\Testing\
Propane Consumption Calculator 7_1_25-7_31_25.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\
7_14_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Bulk\
7_18_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Bulk\
7_1_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Bulk\
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7_31_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Bulk\
7_7_25.PDF	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Bulk\
7_23_25.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Cylinders\
7_2_25.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Cylinders\
7_30_25.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Cylinders\
7_9_25.pdf	\Charm BO MRV\Charm BO MRV\Production Period 8\Propane\Transaction Docs\Cylinders\
7_1_25-7_31_25.png	\Charm BO MRV\Charm BO MRV\Production Period 8\Up-Time Data\
Miniforge Pyrolysis Up-Time Reporting.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Up-Time Data\

PP8 Water Usage Log.xlsx	\Charm BO MRV\Charm BO MRV\Production Period 8\Water\
RP Summary August Mini-RP.xlsx	\Injection Lot Data\Injection Lot Data\
5-01 Charm Contents.xlsx	\Injection Lot Data\Injection Lot Data\05-01\
5-01 Ops Notes.xlsx	\Injection Lot Data\Injection Lot Data\05-01\
BATCH 5-01 QOWV and BATCH 2-103 AECN.pdf	\Injection Lot Data\Injection Lot Data\05-01\
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5-01 Scale In and Out.jpg	\Injection Lot Data\Injection Lot Data\05-01\Scale Tickets\
8_10_25. 5-01. Aqueous Scale Out Final.jpg	\Injection Lot Data\Injection Lot Data\05-01\Scale Tickets\
2-103 Ops Notes.xlsx	\Injection Lot Data\Injection Lot Data\2-103\
BATCH 5-01 QOWV and BATCH 2-103 AECN.pdf	\Injection Lot Data\Injection Lot Data\2-103\
Empty Trailer 3.jpeg	\Injection Lot Data\Injection Lot Data\2-103\Scale Tickets\
Empty Weight 1.jpg	\Injection Lot Data\Injection Lot Data\2-103\Scale Tickets\
Empty Weight 2.pdf	\Injection Lot Data\Injection Lot Data\2-103\Scale Tickets\
Full Weight Trailer 1.pdf	\Injection Lot Data\Injection Lot Data\2-103\Scale Tickets\
Full Weights (2 Trailers).jpg	\Injection Lot Data\Injection Lot Data\2-103\Scale Tickets\
3-40 Charm Contents.xlsx	\Injection Lot Data\Injection Lot Data\3-40\
3-40 Ops Notes.xlsx	\Injection Lot Data\Injection Lot Data\3-40\
Batch 3-40, 3-41, 3-42.PDF	\Injection Lot Data\Injection Lot Data\3-40\
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Scale Ticket Out.jpg	\Injection Lot Data\Injection Lot Data\3-40\Scale Tickets\
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3-42 Charm Contents.xlsx	\Injection Lot Data\Injection Lot Data\3-42\
3-42 Ops Notes.xlsx	\Injection Lot Data\Injection Lot Data\3-42\
Batch 3-40, 3-41, 3-42.PDF	\Injection Lot Data\Injection Lot Data\3-42\
3-42 Charm Contents.xlsx	\Injection Lot Data\Injection Lot Data\3-42\Deprecated\
3-42 Ops Notes.xlsx	\Injection Lot Data\Injection Lot Data\3-42\Deprecated\
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8_1-8_14 Diesel Consumption Calcs.xlsx	\Site Emissions\Site Emissions\Basco Injection\Diesel\
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B6 Electricity Daily Usage.xlsx	\Site Emissions\Site Emissions\Basco Injection\Electricity\
8_1-8_14 Gasoline Tracker KS.xlsx	\Site Emissions\Site Emissions\Basco Injection\Gasoline\
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8_1-8_14 Diesel Consumption Calcs.xlsx	\Site Emissions\Site Emissions\KS Pre-Processing\Diesel\
Charm Industrial INC_csv_2025-08-18_19_54_31.xlsx	\Site Emissions\Site Emissions\KS Pre-Processing\Electricity\

Minifinery 8_1_25-8_14_25 Consolidated Energy Consumption.xlsx	\Site Emissions\Site Emissions\KS Pre-Processing\Electricity\
8_1-8_14 Gasoline Tracker KS.xlsx	\Site Emissions\Site Emissions\KS Pre-Processing\Gasoline\
8_7_25.jpg	\Site Emissions\Site Emissions\KS Pre-Processing\Gasoline\
Basco 6 + KS Embodied Emissions Inventory.xlsx	\Site Emissions\Site Emissions\KS_LA EE\
2035 Gal Agspray Tank.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
250 Gallon Diesel Fuel Tank.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
400 Gallon Diesel Fuel Tank.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
5000 Gallon Tank.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
64x24 Double Wide Office Trailer.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJs CAT 289D3.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJs CAT TL1055D.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJs Chiller EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJs Concrete Pad EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJs Genie S-45 XC EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Ajs Pumps EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJs Thermal Ox EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJ_s IBC Tote 275 Gal.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJ_s IBC Totes 330 Gal.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJ_s Sparge System EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJ_s Tank Piping Upgrade EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
AJ_s Tanks EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Auger System EE.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Ballymore Ladder.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
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Electrical Infrastructure Installation and Upgrades.docx (1).pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
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Fabricated Steel Hopper.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Fuel Cube.docx	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Funk and Steel EE Details - Funk Detail.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Hotsy Pressure Washer.docx	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Indiana Scale Company IN7620.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
KS May 25 Aggregates Calc Sheet - Sheet1.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
KS May 25 Concrete Pads - Sheet1.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
KS May 25 Road Aggregates Calc Sheet - Sheet1 (1).pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
LCS Tank.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
PD Blowers 65URAI.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Pre-Fab Shed.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
ProBox.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Raw Material - Steel.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\

Salt Auger Dosing Motor.docx	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Shipping Container LCA.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
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Sliding Vane Pump.docx	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Steel Dosing Tank.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Steel Fitting and Valve Upgrades.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Storage Tank Base.docx	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Storage Tank.docx	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
TP and M System.png	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
Yale Forklift.docx.pdf	\Site Emissions\Site Emissions\KS_LA EE\KS LCA Docs\
10500 Gal Water Tank.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
40_x8_ Conex Shipping Container.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
500bbl Frac Tank.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
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Centrifugal Recirc Pump.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Diaphragm Seal for Pressure Transducers.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Downhole Monitoring Assembly.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Generac GP2500i.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
JCB Telehandler 512-56.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Kubota L2501 Tractor.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
PC Pump Motor+VFD.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Progressive Cavity Pump.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Site Work Truck.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Spill Containment.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
SRO System.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Steel Bio-Oil Tanks.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Storage Tank Containment.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Sullair Air Compressor.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
Ultrasonic Flow Meter.docx	\Site Emissions\Site Emissions\KS_LA EE\LA LCA Docs\
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CHARM INDUSTRIAL FIGURES.pdf	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\
Charm Louisiana LLC - SN 974025 - Admin Complete Packet.pdf	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\
CURRENT WBD 01.28.25.pdf	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\
fill analysis.pdf	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\
Formation Fluid Characterization --Basco 6 SWD MRVPDFreport_far041-042.PDF	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\
LDENR OW Progress Report 10_2024.pdf	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\

Rpt_2403519_NB_Wood_Chip_Final_v3.pdf	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\
Site Location-Well Area Topography.pdf	\Well Permitting-Characterization-Baseline Scenario-20250919T201945Z-1-001\Well Permitting-Characterization-Baseline Scenario\

APPENDIX 2: VERIFICATION AUDIT FINDINGS AND RESPONSES

ID No	Type	Finding / Issue	Audit Evidence	Required Action	Supplier Response	Conclusion / Resolution
1	Nonconformity - Omission	Weigh Scale ticket for 5-01 injection batch Aqueous delivery is not provided (empty scale ticket is provided, but not full)	8_10_25. 5-01. Aqueous Scale Out Final	Provide clarification or correct info (see Finding 2)	The full scale ticket for 5-01 is included — The original file name was "5-01 Scale In and Out" but was revised to "5-01 Scale In and Pre-load empty" for improved clarity as there are two scale tickets in the image: The pre-loaded empty mass (33600 lbs, weigh number 1408) and the full mass (61200 lbs, weigh number 2539). The pre-load empty is taken so there is a redundant unloaded datapoint, but the post-unload mass is preferred, so it is not used unless the post-unload mass is not available. The full mass reflects the same value that was included in the original ops notes and calculations for this removal.	Thank you for the clarification. Complete.
2	Nonconformity - Error	It appears that the wrong empty weight was used for the bio-oil shipment truck weight for the 5-01 delivery of bio-oil. The weigh scale ticket shows 33600, but MRV Ops notes file shows 31620 (as does Certify - removal YP7J) - which is the aqueous truck empty weight and for a different truck and trailer. This drops the mass of bio-oil injected by 898 kg when corrected. Ensure that the total gross removals value which is used elsewhere is corrected and carried through (not material): Total embodied emissions (+2kg additional); transport emissions (empty truck) = -67kg additional; truck embodied (negligible)	8_10_25. 5-01. Aqueous Scale Out Final 5-01 Scale In and Out	Provide correct final weigh ticket for 05-01	The scale ticket showing 31620 lbs was included in error and is not the correct post-unload empty weight. The correct ticket providing the post-unload empty weight was identified and confirmed that the correct date and tractor/trailer combination are listed. The ticket weight is 33480 lbs and the weigh number is 3159. Ops notes and calculations in Certify have been updated to reflect the corrected value. This correction reduced the MOBI for Batch 5-01 by 843.68kg and reduced to total reported CDR by 280kg which is reflected in an updated GHG Statement Report.	Complete. GHG Statement modified 19SEP to account for updated weigh ticket.
3	Clarification (potential	It does not appear the rental injection pump observed on site was included in the embodied emissions LCA, unless that is included within the calculation	Basco 6 + KS Embodied	Add rental pump to embodied emissions	The rental pump was not put into service until after 8/15. Site injections for the first week were executed using the Continental pumps. The rental pump was put into service after	Complete. Thanks for the clarification.

	omission - not material)	for the 3 Continental well pumps on the Basco 6 + KS Embodied Emissions Inventory.xls sheet (Not material)	Emissions Inventory	equipment list and account for	those pumps began showing performance issues. The rental pump will be included in the EE inventory for the next Reporting Period	
4	Clarification	Are the tote tanks used on site accounted for in the prior charm embodied emissions calculations and equipment lists? They don't appear in the embodied equipment inventory for Basco Basco 6 + KS Embodied Emissions Inventory.xls	Basco 6 + KS Embodied Emissions Inventory	Clarify if tote tanks at Basco are included in list of equipment from CO/KS	The totes come from and return to CO, so they are part of the CO inventory.	Complete. Thanks for the clarification.
5	FAR (prior)	The PDD should be updated to reflect GHG Statement calculation changes as noted in the referenced briefs				PDD Updated. Complete
6	FAR (prior)	Please plan to provide updated documentation on community engagement and stakeholder input activities by 12/31/2025			Will do -- Working with our Community Engagement Manager on this item	Review for future verifications
7	FAR (prior)	Please plan to provide updated documentation on grievance mechanism updates and any complaints/grievances received and responses (or confirmation of no issues) by 12/31/2025			Will do -- Working with our Community Engagement Manager on this item	Review for future verifications
8	FAR (prior)	Please ensure all waste disposal related emissions for future disposal of wastes which have been produced during this period are accounted for in future reporting periods as required. Note: Charm has properly accounted for all disposals so far and has all systems in place to do so going forward.			Absolutely -- We will continue to account for completed disposals as soon as they are completed and, if applicable, documentation is provided to quantify mass, confirm disposal method, etc.	Not applicable for current verification. Review for future verifications

9	Nonconformity	A description of the confining system or caprock is not complete. Although indications are that the system holds significantly more pressure than planned injection MASIP, and injections will be well below the USDW lowest point, details regarding formation and faults or transmissivity are unclear		Please provide a summary of the confining system and details, including	This is addressed in the document titled "Basco 6 Supplementary Subsurface Characterization" which is now included in the "VVB Feedback" subfolder of the "Additional Documentation" folder in addition to supporting evidence. The text of the document, minus attachment links that won't work for the public, is also reproduced as Appendix 9 of the updated PDD	Complete. Additional evidence supporting presence of confining layers provided in 'Basco 6 EW L Miocene Verification Requests.docx'. Note - if future reperforations or injection well modifications are planned for injection at shallower depths, we would request similar data to the well bore GR, GR KUT, and other test results be provided for review and as evidence of well construction, reservoir characteristics, and confining layers in the planned injection region.
10	Clarification	Did not see DIC or carbon isotope test results for formation fluid nor hydrocarbon quantification for baseline fluid	This is also addressed in the Basco 6 Supplementary Subsurface Characterization using results from the formation test also included in the same folder identified in NC 9	Please provide or direct to any results for DIC testing and hydrocarbon analysis, and, if available, carbon isotope testing (not required)	This is also addressed in the Basco 6 Supplementary Subsurface Characterization using results from the formation test also included in the same folder identified in NC 9	Complete. Baseline DIC, C and O isotope test results provided in Job59299-Water.pdf and 'Basco 6 EW L Miocene Verification Requests.docx'. Please provide results of any future characterizations as part of that reporting period data package.
11	Nonconformity	Reservoir mineralogy and lithology descriptions are minimal	Additional mineralogy and lithology descriptions are also included in the Basco 6 Supplementary Subsurface Characterization document	Please provide or direct to any more detailed descriptions of formation beyond the brief summary in the PDD	Additional mineralogy and lithology descriptions are also included in the Basco 6 Supplementary Subsurface Characterization document	Complete

12	FAR	Site Formation characterization is required to be reviewed every ten years or when site conditions warrant, with comparison of pre-emplacement project assumptions to actual measured conditions and updates as needed		Please add the 10 yr review and review upon collection of measured conditions or observations differing from initial review to the Permit Compliance Plan in the Supplemental Document - Basco 6	This has been added to the Permit Compliance Plan	Complete
13	Nonconformity	Protocols require validation that well materials designed for life of project and compatible with fluids / injectate and meet API, ASTM, ISO or comparable standards as applicable (i.e. corrosion resistance). Although well materials appear to be designed properly and are compatible, no evidence is provided on materials used not compliance with ASTM, API, or other standards		Please provide supporting information on construction materials used and required standards	This information is provided in the document titled "Material Selection and Service Life - Basco 6 EW 001" in the "VVB Feedback" subfolder of the "Additional Documentation" folder in addition to supporting evidence	Complete
14	Clarification	Carbon, oxygen, and hydrogen isotope signature analysis of formation fluid (and bio-oil) is recommended by the protocol. Has any been completed or is planned?		Please clarify if this testing is planned or has been completed. Note it is not fully required by the protocol.	The isotope signature analyses referenced in and included as evidence alongside the Supplemental Subsurface Characterization doc are the only isotope signature analyses that are planned to be completed at this time.	Complete

APPENDIX 3: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	Criteria Met?	Evidence / Notes *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. The majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Steven Qiu

Carbon Removal Verification Engineer, 350Solutions

EDUCATION:

MSc Carbon Management, University of Edinburgh, UK, 2024
BSc Economics, Dalhousie University, Canada, 2018

EXPERIENCE SUMMARY:

At 350 Solutions, Steven's efforts center on validation and verification of varied carbon removal pathways. Steven began his experience in the carbon markets in 2023 providing technical due diligence for Carbon Direct on a BECCS project in Canada. Then, following completion of the MSc Carbon Management program at the University of Edinburgh, he continued finding work with leading CDR and climate organizations. Patch, a CDR marketplace, has contracted Steven as a Carbon Project Diligence Scientist where he provides scientific review services for a wide range of carbon project types. Pembina Institute, a leader in sustainability policy and research in Canada, works with Steven to publish novel research on Canadian carbon removal.

RELEVANT PROFESSIONAL EXPERIENCE:

Oct 2024 – Present: Carbon Removal Verification Engineer, 350Solutions
Verify carbon dioxide removal technologies on behalf of registries and private companies ensuring high quality and meaningful climate impact.
Sept 2024 – Present: Carbon Project Diligence Scientist, Patch
Conduct deep scientific diligence on carbon projects (biochar, IFM, REDD+, ARR, and others) according to Patch's Project Acceptance Criteria to build trust in the integrity of climate solutions.
Nov 2024 – Present: Carbon Removal Expert, Pembina Institute
Research topics relevant to carbon removal policy in Canada.
June 2023 – Present: Scientific Project Reviewer, Carbon Direct